

# DEV PATEL

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## OVERVIEW

Software Engineering graduate student passionate about building scalable web applications and intelligent AI-powered systems using full-stack development, backend engineering, and modern Retrieval-Augmented Generation (RAG) technologies.

## PROFESSIONAL SUMMARY

Graduate student pursuing an M.S. in Computer Science at The University of Texas at Arlington with hands-on experience in software development using Python, ASP.NET Core, Angular, SQL, and REST APIs. Experienced in developing scalable backend systems, database-driven applications, and enterprise web solutions in Agile environments. Recently expanded into Artificial Intelligence and Machine Learning through projects involving Retrieval-Augmented Generation (RAG), Hugging Face APIs, vector databases, and intelligent document-based systems. Strong interest in Software Engineering, Machine Learning, Generative AI, and scalable AI-driven applications with a passion for solving real-world problems through technology.

## TECHNICAL SKILLS

- Programming Languages:** C#, Python, TypeScript, JavaScript, SQL, PowerShell
- Frameworks & Web Technologies:** ASP.NET Core, Angular, RESTful APIs, Web APIs, HTML5, CSS3, JSON, RDLC Reports
- Databases & Data Engineering:** MySQL, SQL Server, LINQ, Stored Procedures, Query Optimization, Data Modeling
- Artificial Intelligence & Data Science:** Retrieval-Augmented Generation (RAG), Generative AI, Hugging Face APIs, Vector Databases (Pinecone), Machine Learning Fundamentals, Computer Vision Basics, Data Preprocessing, Feature Engineering, Model Evaluation
- Software Engineering Practices:** Object-Oriented Programming (OOP), MVC Design Pattern, API Design, Error Handling
- Cloud, DevOps & Tooling:** Git, GitHub, CI/CD Pipelines, Vercel, Render, Azure DevOps, PowerShell Automation, VS Code
- Testing & Quality Assurance:** Unit Testing, API Testing (Postman), Debugging, Code Reviews
- Development Methodologies:** Agile/Scrum, Requirement Analysis, Cross-functional Collaboration

## PROFESSIONAL EXPERIENCE

### Software Developer (.NET)

Dev Information Technology Limited, Jaipur, India | Jun 2024 – Jun 2025

- Designed, developed, and maintained enterprise-scale web applications using ASP.NET Core and Angular.
- Built and consumed RESTful APIs to support modular, scalable backend services.
- Optimized MySQL queries, stored procedures, and indexing, improving application performance and response times.
- Developed RDLC reports for operational insights and data-driven decision-making.
- Worked in an Agile/Scrum environment, participating in sprint planning, daily standups, and code reviews.
- Collaborated with frontend, QA, and product teams to deliver production-ready features.

### Software Developer Intern

Dev Information Technology Limited, Jaipur, India | Jan 2024 – May 2024

- Assisted in backend business logic implementation and responsive UI development using .NET and Angular.
- Supported API development, debugging, and testing for existing enterprise modules.
- Contributed to database enhancements and feature releases under senior developer guidance.
- Gained exposure to real-world production systems, version control, and CI/CD workflows.

## EDUCATION

Master of Science (M.S.) — Computer Science | The University of Texas at Arlington | Arlington, TX | 2025 – 2027 (*Expected*)

**Relevant Coursework:** Advanced Software Engineering, Data Structures & Algorithms, Artificial Intelligence, Computer Vision, Data Mining, Cloud Computing & Big Data (CCBD), Database Systems, Big Data Analytics

Bachelor of Technology (B.Tech) — Information Technology | Ganpat University | Gujarat, India | 2020 – 2024

**Relevant Coursework:** Object-Oriented Programming (OOP), Programming for Problem Solving (PPS), Cloud Computing, Artificial Intelligence, Database Management Systems, Web Technologies, Operating Systems, Computer Networks

## PROJECTS

AI-Powered RAG Document Assistant | Next.js, FastAPI, Python, Hugging Face, Pinecone | [Link](#)

- Built and deployed a full-stack AI-powered document assistant that allows users to upload PDF/TXT files and ask natural-language questions from document content.
- Implemented Retrieval-Augmented Generation (RAG) concepts including document parsing, text chunking, semantic retrieval, and AI-assisted answer generation.
- Developed RESTful backend APIs using FastAPI and designed a responsive frontend using Next.js and TypeScript.
- Integrated Hugging Face APIs and Pinecone vector database for document retrieval and intelligent response generation.
- Deployed the frontend on Vercel and backend on Render with GitHub-based CI/CD workflow for production-ready hosting.

Multi-View Geometry: Classical vs Deep Feature Matching

- Benchmarked classical and deep-learning-based feature matching approaches including SIFT, ORB, SuperPoint, and LightGlue on the KITTI autonomous driving dataset across 5,600+ image frames.
- Implemented feature detection, descriptor extraction, and robust correspondence matching pipelines using Python and OpenCV.
- Evaluated geometric consistency through fundamental matrix estimation, epipolar constraints, RANSAC filtering, and camera pose recovery.
- Compared accuracy, robustness, and matching performance of traditional computer vision methods against modern deep feature matching approaches in real-world driving scenarios.

Kaushal Darpan – College Management System | Government of Rajasthan, India.

- Built a large-scale college management system using ASP.NET Core and Angular.
- Developed modules for student records, fee management, reporting, and departmental workflows.
- Built secure REST APIs and optimized SQL queries, improving data retrieval performance by ~30%.

NICSI Project – Government IT Services Platform | Government of India.

- Contributed to a government IT services platform under NICSI initiatives.
- Worked on backend development, REST APIs, and SQL-based data operations using ASP.NET Core.
- Supported reporting, validation, and secure application workflows aligned with public-sector standards.

## CERTIFICATIONS

- [Complete C# Masterclass \(Udemy\)](#)
- [Virtual Internship Program \(TechnoHacks EduTech\)](#) [Python And Machine Learning \(DevTown\)](#)
- [Cybersecurity Analyst Job Simulation](#)